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UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: TEJS 228

COURSE TITLE: TEACHING AND ASSESSING JHS MATHEMATICS (INTRO)

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A

[25 marks]

Answer all the questions in this section.

1. Which of the following best explains why a standards-based curriculum might be more responsive to learner diversity than an objective-based curriculum?
   A. Standards-based curricula provide a broader range of content to assess students on multiple subjects equally.
   B. Objectives are often more measurable than standards, which makes them less flexible in diverse classrooms.
   C. Standards emphasize what learners should be able to do, allowing differentiated instruction aligned with competencies.
   D. Objective-based curricula rely on behavioral outcomes, which makes them easier to modify for different learners.

2. A teacher wants to teach subtraction of numbers within 999 using the standards-based curriculum. Which approach best aligns with the curriculum's philosophical orientation?
   A. Drill exercises on column subtraction with and without regrouping.
   B. Use of manipulatives and group activities to solve real-life subtraction problems.
   C. Assign individual tests after explaining the steps in detail.
   D. Teach the rules of subtraction first, then model problems on the board.

3. Which of the following is a fundamental distinction between standards-based and objective-based curricula in mathematics?
   A. Standards are less precise than objectives, making them unsuitable for assessment.
   B. Objectives define broad competencies, whereas standards focus on observable behaviors.
   C. Standards reflect long-term goals and skills development, while objectives often target short-term mastery.
   D. Objectives are student-centered, while standards are teacher-centered.

4. During a lesson on number sense, a student insists that 0.5 is greater than 0.73 because "5 is greater than 3." How should a teacher using a constructivist approach respond?
   A. Correct the student immediately and give a new example.
   B. Provide the correct answer and ask others to explain why.
   C. Use manipulatives like fraction strips to allow the student to test their assumption.
   D. Tell the student to revise decimals at home before continuing.

5. Which pair of teaching strategies best demonstrates the relationship between multiplication and division in a JHS context?
   A. Rote learning of multiplication facts and timed division drills.
   B. Modeling repeated addition for multiplication and repeated subtraction for division.
   C. Group work on prime factorization and multiplication of decimals.
   D. Demonstrating commutativity for both operations using story problems.

6. In designing an assessment tool aligned with the standards-based curriculum, which of the following steps would most likely violate good assessment practice?
   A. Defining the purpose and format before writing questions.
   B. Using only multiple-choice items to test reasoning and communication.
   C. Creating a test blueprint with varied difficulty levels.
   D. Including rubrics for open-ended tasks.

7. Which classroom scenario best reflects "assessment as learning" (AaL) in mathematics?
   A. The teacher reviews students' end-of-unit test scores and provides feedback.
   B. Students complete an exit ticket quiz to show what they've learned that day.
   C. Learners use peer feedback to adjust their strategies while solving problems.
   D. The teacher conducts oral questioning to revise the previous lesson.

8. A student confidently solves $45 \div 5$ as 5, explaining that "5 goes into 45 nine times, so the answer must be 5." What does this error suggest about their mathematical thinking?
   A. They are applying the inverse operation incorrectly and confusing dividend and divisor.
   B. They misunderstood the operation because they memorized the wrong fact.
   C. They correctly applied repeated subtraction but forgot to subtract.
   D. They are confusing multiplication properties with division properties.

9. In a lesson using technology for teaching fractions, which of the following is the most effective strategy to support conceptual understanding?
   A. Showing an animation of fraction addition and asking students to take notes.
   B. Using an online game that awards points for correct answers on equivalent fractions.
   C. Engaging students with virtual manipulatives to explore part-whole relationships.
   D. Projecting pre-solved problems for students to copy and revise.

10. Which of the following best captures the aim of mathematics education in Ghana's basic schools according to the standards-based curriculum?
   A. To prepare learners to pass the BECE with high proficiency.
   B. To develop learners' abilities to memorize formulas and rules.
   C. To empower learners to apply mathematical thinking in solving real-world problems.
   D. To encourage learners to practice mathematics for personal enjoyment only.

11. Ama says, "When I subtract a number from 10,000 and then subtract the result from 10,000 again, I always get back the original number." Which number will break Ama's pattern?
    A. 4,321
    B. 10,000
    C. 0
    D. 5,678

12. A teacher writes the number 40,062 on the board and asks, "Which digit is in the position that is ten times the value of the digit 6?" What is the correct answer?
    A. 0
    B. 4
    C. 2
    D. 6

13. A farmer harvested 2,463 oranges and packed them equally into boxes of 9. He mistakenly reports that he used 270 boxes. Which of the following best evaluates the accuracy of his report?
    A. The report is accurate because 2,463 is divisible by 9.
    B. The report is inaccurate; 2,463 ÷ 9 is not 270 but 273.
    C. The report is inaccurate; 2,463 ÷ 9 gives a remainder.
    D. The report is accurate if we assume some oranges were discarded.

14. A number increases by 25%, then decreases by 20%. What is the overall percentage change?
    A. 45%
    B. 5% decrease
    C. 5% increase
    D. 20%

15. "Multiply 49 by 5, then divide the result by 10."
    Which of the following strategies would yield the same answer more efficiently?
    A. Multiply 49 by 0.5
    B. Multiply 49 by 0.25
    C. Divide 49 by 2
    D. Multiply 49 by 0.5, then by 1

16. I am a 3-digit number. My tens digit is 3 more than my units digit. My hundreds digit is twice the units digit. What number am I?
    A. 863
    B. 942
    C. 462
    D. 381

17. Which of the following is the correct interpretation of the fraction $\frac{5}{8}$ in a part-whole model using a number line?
    A. The fifth mark after dividing the number line from 0 to 1 into 8 equal parts
    B. The mark halfway between 0 and 8 on a whole-number line
    C. A line segment starting at 5 and ending at 8
    D. One of 5 equal parts taken from a group of 8

18. A teacher is designing an activity to help learners understand equivalent fractions. Which task best supports this aim?
    A. Ask students to write five fractions with different numerators
    B. Have learners fold paper strips into equal parts and shade equivalent sections
    C. Give learners a test on simplifying fractions
    D. Use a number line to identify prime numbers between 0 and 1

19. Which of the following problems is best suited for introducing the idea of negative integers in a relatable way?
    A. A farmer harvests 50 bags of maize
    B. A student scores 10 out of 10 on a test
    C. A diver goes 20 meters below sea level
    D. A clock shows 3 o'clock in the afternoon

20. A student calculates: "$-4 - (-7) = -11$". What does this reveal about the student's understanding?
    A. The student subtracted the numbers without recognizing the double negative
    B. The student added the numbers but forgot to include the sign
    C. The student misunderstood the associative property
    D. The student applied multiplication rules to subtraction

21. Which of the following number sentences correctly demonstrates the distributive property of integers over addition?
    A. $-3 \times (2+4) = -3 \times 2 + 4$
    B. $-3 \times (2+4) = -6 + (-12)$
    C. $-3 \times (2+4) = -6 + (-12)$
    D. $-3 \times (2+4) = -6 + (-6)$

22. Which system is emphasized for teaching place value in the curriculum?
    A. Binary number system
    B. Base-ten number system
    C. Hexadecimal system
    D. Octal number system

23. In the curriculum, multiplication is demonstrated as:
    A. Division in reverse only
    B. Repeated addition and arrays of numbers
    C. Only through mathematical formulas
    D. Abstract concepts without manipulatives

24. Which assessment type focuses on gathering information during the learning process?
    A. Assessment of Learning (AoL)
    B. Assessment for Learning (AfL)
    C. Summative Assessment only
    D. Final Evaluation

25. In the context of number systems, what is the primary purpose of using manipulatives in teaching place value?
    A. To make lessons more entertaining
    B. To demonstrate abstract concepts concretely
    C. To reduce teacher workload
    D. To keep students busy

SECTION B

[75 marks]

Answer three questions from this section.

1. a) What is a lesson plan. ***(3 marks)***

   b) In four (4) ways, explain why lesson planning is important. ***(4 marks each)***

   c) A farmer has 240 acres of land. He plants: $\frac{1}{4}$ of the land with maize, $\frac{1}{3}$ with rice, and the rest with cassava.

   (i) What fraction of the land is used for cassava? ***(2 marks)***

   (ii) What is the percentage of fraction used for cassava? ***(2 marks)***

   (iii) Determine the acre(s) of land used for cassava. ***(2 marks)***

2. Discuss **FIVE** (5) roles of mental strategies in developing computational fluency in basic arithmetic operations. ***(25 marks)***

3. Evaluate **FIVE** (5) roles of feedback and learner self-assessment in improving learning outcomes. ***(25 marks)***

4. a) Distinguish among Assessment for Learning (AfL), Assessment of Learning (AoL), and Assessment as Learning (AaL). Give two (2) examples each. ***(16 marks)***

   b) Explain assessment tools and give three examples. ***(9 marks)***

5. a) A water tank is in the shape of a cylinder with a height of 10 m and a radius of 5 m.

   (i) Find the volume of the tank. ***(6 marks)***

   (ii) If the tank is half full, what is the volume of water inside? ***(3 marks)***

   b) Find the product of the following, using the Lattice Method

   (i) 7285 and 3212 ***(6 marks)***

   (ii) 605 and 493 ***(4 marks)***

   (iii) 3731 and 4041***(6 marks)***